Question # 8 of 10 (Start time: 09:58:04 AM, 04 January 2022) Total Marks: 1

An annihilator operator L for the function, $f(x) = e^{3x}$, is $L =$ _____.

Select the correct option

Reload Math Equations

○	e^{3x}
●	$D - 3$
○	$D - e^{3x}$
○	$3e^{3x}$

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 09:40:39 AM, 04 January 2022) **Total Marks:** 1

If

$$y = e^{5x}$$

, then which of the following is true for it?

Select the correct option

Reload Math Equations

○	Its annihilator operator is $(D+5)^2$
○	Its annihilator operator is (D+5)
○	Its annihilator operator is $(D-5)^2$
●	Its annihilator operator is (D-5)

Click to Save Answer & Move to Next Question

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Pdf.pdf 151 / 448 100%

Trial particular solutions

Number	The input function $g(x)$	The assumed particular solution y_p
1	Any constant e.g. 1	A
2	$7x+7$	$Ax+B$
3	$3x^2-2$	Ax^2+Bx+c
4	x^3-x+1	Ax^3+Bx^2+Cx+D
5	$\sin 4x$	$A\cos 4x+B\sin 4x$
6	$\cos 4x$	$A\cos 4x+B\sin 4x$
7	e^{5x}	Ae^{5x}
8	$(9x-2)e^{5x}$	$(Ax+B)e^{5x}$
9	x^2e^{5x}	$(Ax^2+Bx+C)e^{5x}$
10	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x+Be^{3x}\sin 4x$
11	$5x^2\sin 4x$	$(A_1x^2+B_1x+C_1)\cos 4x+(A_2x^2+B_2x+C_2)\sin 4x$
12	$xe^{3x}\cos 4x$	$(Ax+B)e^{3x}\cos 4x+(Cx+D)e^{3x}\sin 4x$

Trial particular solutions

ber	The input function $g(x)$	The assumed particular solution y_p
	Any constant e.g. 1	A
	$5x+7$	$Ax+B$
	$3x^2-2$	Ax^2+Bx+c
	x^3-x+1	Ax^3+Bx^2+Cx+D
	$\sin 4x$	$\cos 4x + B\sin 4x$
	$\cos 4x$	$A\cos 4x + B\sin 4x$
	e^{5x}	Ae^{5x}
7	$(9x-2)e^{5x}$	$(Ax+B)e^{5x}$
8	x^2e^{5x}	$(Ax^2+Bx+C)e^{5x}$
9	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x + Be^{3x}\sin 4x$
10	$5x^2\sin 4x$	$(A_1x^2+B_1x+C_1)\cos 4x+(A_2x^2+B_2x+C_2)\sin 4x$
11	$xe^{3x}\cos 4x$	$(Ax+B)e^{3x}\cos 4x + (Cx+D)e^{3x}\sin 4x$
12		

	The input function $g(x)$	
	Any constant e.g. 1	A
	$5x+7$	$Ax+B$
	$3x^2-2$	Ax^2+Bx+c
	x^3-x+1	Ax^3+Bx^2+Cx+D
	$\sin 4x$	$A\cos 4x+B\sin 4x$
	$\cos 4x$	$A\cos 4x+B\sin 4x$
	e^{5x}	Ae^{5x}
	$(9x-2)e^{5x}$	$(Ax+B)e^{5x}$
	x^2e^{5x}	$(Ax^2+Bx+C)e^{5x}$
10	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x+B$ $\sin 4x$
11	$5x^2\sin 4x$	$(A_1x^2+B_1x+C_1)\cos 4x+(A_2x^2+B_2x+C_2)\sin 4x$
12	$xe^{3x}\cos 4x$	$(Ax+B)e^{3x}\cos 4x+(Cx+D)e^{3x}\sin 4x$

17°C

$$(D-a)^n$$

nihilates each of the functions

$$e^{ax}, xe^{ax}, x^2e^{ax}, \ldots, x^{n-1}e^{ax}$$

Hence, as a consequence of the fact that the differentiation can be
erm, the differential operator

$$(D-a)^n$$

annihilates the function

$$y = c_1e^{ax} + c_2xe^{ax} + c_3x^2e^{ax} + \cdots + c_nx^{n-1}e^{ax}$$

Example 8

Find an annihilator operator for the functions

(a) $f(x) = e^{5x}$

(b) $g(x) = 4e^{2x} - 6xe^{2x}$

Solution

(a) Since $(D-5)e^{5x} = 5e^{5x} - 5e^{5x} = 0.$

Therefore, the annihilator operator of function f is given by

$$L = D - 5$$

We notice that in this case $a = 5,\ n = 1.$

(b) Similarly $(D-2)^2(4e^{2x} - 6xe^{2x}) = (D^2 - 4D + 4)(4e^{2x}) - (D^2 - 4D + 4)(6x$

$n-1$ derivatives of y_p. The last equation in the system results from substituting the particular integral y_p and its derivatives into the given nth order linear differential equation and then simplifying.

Depending upon how the integrals of the derivatives u_k' of the unknown functions are found, the answer for y_p may be different for [illegible] to find y_p for the same equation.

When asked to solve an initial value problem, we need to be sure to apply the initial conditions to the general solution and not to the complementary function alone, thinking that it is only y_c that involves the arbitrary constants.

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Trial particular solutions

er	The input function $g(x)$	The assumed particular solution y_p
	Any constant e.g. 1	A
	$5x+7$	$Ax+B$
	$3x^2-2$	Ax^2+Bx+c
	x^3-x+1	Ax^3+Bx^2+Cx+D
	$\sin 4x$	$A\cos 4x+B\sin 4x$
	$\cos 4x$	$A\cos 4x+B\sin 4x$
	e^{5x}	Ae^{5x}
	$(9x-2)e^{5x}$	$(Ax+B)e^{5x}$
		$(Ax^2+Bx+C)e^{-x}$
	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x+Be^{3x}\sin 4x$
11		$(A_2x^2+B_2x+C_2)\cos 4x+(A_1x^2+B_1x+C_1)\sin 4x$
12	$xe^{3x}\cos 4x$	$(Ax+B)e^{3x}\cos 4x+(Cx+D)e^{3x}\sin 4x$

If $g(x)$ equals a sum?

Suppose that

Question # 1 of 10 (Start time: 09:18:52 AM, 04 January 2022)

An annihilator operator L for the function, $f(x) = e^{3x}$, is $L =$ _____.

Select the correct option

Reload

○	$D - e^{3x}$
○	$3e^{3x}$
○	e^{3x}
◉	$D - 3$

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MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 09:18 AM, 04 January 2022

Question # 2 of 10 (Start time: 09:19:21 AM, 04 January 2022)

Total Marks: 1

What is annihilator operator of the function

$$g(x) = x^4 + 5$$

?

Select the correct option

Reload Math Equations

○	(D+4)
○	(D-4)
◉	D^5
○	$(D+5)^4$

Click to Save Answer & Move to Next Question

Question # 3 of 10 (start time: 09:19:47 AM, 04 January 2022) Total Marks: 1

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Select the correct option

Reload Math Equations

○	D^2
○	$D+5$
●	$D+1$
○	D

Click to Save Answer & Move to Next Question

Question # 3 of 10 (start time: 09:19:47 AM, 04 January 2022) Total Marks: 1

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Select the correct option

Reload Math Equations

○	D^2
○	$D+5$
●	$D+1$
○	D

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 09:20:57 AM, 04 January 2022) **Total Marks:** 1

If

$$y = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x}$$

is the complementary solution of

$$y'' + 4y' - 2y = 2x^2 - 3x + 6$$

,then the general form of its particular solution is

$$y_p$$

=_____.

Select the correct option

Reload Math Equations

○	$Ax^2 + Bx + C$
○	$(Ax^2 + Bx)2x^2$
○	$Ax + B$
○	$Ax^3 + Bx^2 + Cx + D$

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Question # 4 of 10 (Start time: 09:20:15 AM, 04 January 2022) **Total Marks:** 1

If

$$y = c_1 e^x + c_2 e^{4x}$$

is the complementary solution of

$$y'' - 5y' + 4y = 8e^x$$

, then the general form of its particular solution is

$$y_p$$

=_____.

Select the correct option

Reload Math Equations

◉	Axe^x
○	Ae^x
○	None of these
○	$(Ax + B)e^{4x}$

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 09:20:36 AM, 04 January 2022) **Total Marks:** 1

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

Reload Math Equations

○	5D
○	D+1
○	D+5
◉	D

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Question # 7 of 10 (Start time: 09:22:22 AM, 04 January 2022)

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

- ○ distributive
- ○ identical
- ○ inverse
- ◉ commute

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$(Ax + B)e^{5x}$
○	$Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
●	Ae^{5x}
○	$(Ax^2 + Bx + C)e^{5x}$

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 09:22:37 AM, 04 January 2022) **Total Marks:** 1

If

$$y = e^{(1/2)x}\left(c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x\right)$$

is the complementary solution of

$$y'' - y' + y = 2\sin 3x$$

, then the general form of its particular solution is

$$y_p$$

=_____.

Select the correct option

Reload Math Equations

○	$A\cos 3x + B\sin 3x$
○	$3Asin3x + 3B\cos 3x$
○	$3\cos 3x + 3\sin 3x$
○	$3A\cos 3x + 3A\sin 3x$

Click to Save Answer & Move to Next Question

Question # 1 of 10 (start time: 09:31:42 AM, 04 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	Ae^{4x}
○	$(A + B)(\cos 4x + \sin 4x)$
●	$A\cos 4x + B\sin 4x$
○	$Ax^2 + Bx + C$

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 09:24:18 AM, 04 January 2022) Total Marks: 1

The input function $g(x)$ can have one of the forms: A constant function k; A polynomial function; _____.

Select the correct option

Reload Math Equations

◉	(IV) All (I), (II) and (III).
○	(I) An exponential function $e^{\alpha x}$
○	(II) The trignometric functions $\sin \beta x$, $\cos \beta x$
○	(III) Finite sums and products of these functions

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Question # 2 of 10 (Start time: 09:32:01 AM, 04 January 2022)

In differential equations, the symbol D is known as _____ operator.

Select the correct option

○	relational
◉	differential
○	arithmetic
○	integral

Cli

Question # 3 of 10 (start time: 09:32:31 AM, 04 January 2022) **Total Marks: 1**

The annihilator operator for the function, $g(x) = 5x$, is _____.

Select the correct option

Reload Math Equations

○	$D+5$
○	$D+1$
●	D^2
○	D

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 09:33:05 AM, 04 January 2022)

The displacement measured below the equilibrium position are _____.

Select the correct option

◉	positive
○	negative

Cli

Question # 6 of 10 (start time: 09:33:43 AM, 04 January 2022)

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

- identical
- commute
- distributive
- inverse

sec(s)

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 09:31 AM, 04 January 2022

Question # 5 of 10 (Start time: 09:33:22 AM, 04 January 2022) Total Marks: 1

What is annihilator operator of the function

$$g(x) = 4\sin x$$

?

Select the correct option

Reload Math Equations

○	$(D^2 - 1)$
●	$(D^2 + 1)$
○	$(D^2 + 4)$
○	$(D^2 - 4)$

Click to Save Answer & Move to Next Question

Question # 7 of 10 (start time: 09:33:57 AM, 04 January 2022) **Total Marks:** 1

If

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option

Reload Math Equations

◉	Its annihilator operator is D^2
○	Its annihilator operator is D+1
○	Its annihilator operator is D
○	Its annihilator operator is D^3

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 09:34:32 AM, 04 January 2022) Total Marks: 1

The variation of parameters consists of finding a function $u_1(x)$ such that, $y_p = u_1(x) + y_1(x)$, is a _____ solution of the non - homogeneous differential equation, $\frac{dy}{dx} + P(x)y = f(x)$.

Select the correct option

Reload Math Equations

- complementary
- exact
- general
- particular

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 09:3

Question # 8 of 10 (Start time: 09:34:17 AM, 04 January 2022)

The equation of free un-damped motion is _____.

Select the correct option

Re

- $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$
- $\frac{d^2x}{dt^2} - \frac{k}{m}x = 0$

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Question # 10 of 10 (Start time: 09:35:01 AM, 04 January 2022)

The hook's law states that the force F is proportional to the _____.

Select the correct option

○	Length
◉	Elongation
○	Weight
○	None of these

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Question # 3 of 10 (Start time: 09:37:43 AM, 04 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x + 7$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$Ax^2 + Bx + C$
○	$Ax^3 + Bx^2 + Cx + D$
○	A
◉	$Ax + B$

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz#2)

Question # 5 of 10 (Start time: 09:38:30 AM, 04 January 2022)

The displacement measured below the equilibrium position are _____.

Select the correct option

- negative
- positive

Cli

Question # 4 of 10 (Start time: 09:38:03 AM, 04 January 2022)

The differential operator L is called an annihilator operator of the function f if: _____.

Select the correct option

○	(ii) y = f(x) defines a sufficiently differentiable function.
○	(iii) The function f is such that L(y)=0.
●	(iv) All (i), (ii) and (iii).
○	(i) L is a linear differential operator with constant coefficients.

Click to Sa

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g$
If the input function, $g(x) = \cos 4x$, then the assumed particular solution y_p could be _____

elect the correct option

Reloa

○	$Ax^2 + Bx + C$
○	Ae^{4x}
●	$A\cos 4x + B\sin 4x$
○	$(A + B)(\cos 4x + \sin 4x)$

Click to Save Answer & Move t

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MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 09:37 AM, 04 January 2022

Question # 8 of 10 (Start time: 09:39:17 AM, 04 January 2022) Total Marks: 1

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = 4$, $W_2 = -\frac{1}{5}$, then $u'_2 =$ _____.

Select the correct option

Reload Math Equations

○	4
○	-20
●	$-\frac{1}{20}$
○	$-\frac{1}{5}$

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Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
○	$(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$
○	$A \cos 4x + B \sin 4x$
●	$(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:16 PM, 03 January 2022

Question # 9 of 10 (Start time: 07:22:37 PM, 03 January 2022)

Total Marks: 1

Wronskian of the functions, $y_1(x) = 1,\ y_2(x) = \sin x,\ y_3(x) = \cos x$, is _____.

Select the correct option

Reload Math Equations

○	2
○	-1
✓	1
○	0

Click to Save Answer & Move to Next Question

We can derive a differential equation governing the motion of a mass attached to spring when the Newton's second law combined with _____

Select the correct option

- Hook's la
- Newton's 3rd law

Click to Save Answer & M

Question # 9 of 10 (Start time: 09:40:12 AM, 04 January 2022)
Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = (9x - 2)e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

◉	$(Ax + B)e^{5x}$
○	Ae^{5x}
○	$(Ax^2 + Bx + C)e^{5x}$
○	$Ae^{5x}\cos 5x + Be^{5x}\sin 5x$

Click to Save Answer & Move to Next Question

Trial particular solutions

ber	The input function $g(x)$	The assumed particular solution y_p
	Any constant e.g. 1	A
	$5x+7$	$Ax+B$
	$3x^2-2$	Ax^2+Bx+C
	x^3-x+1	Ax^3+Bx^2+Cx+D
	$\sin 4x$	$A\cos 4x+B\sin 4x$
	$\cos 4x$	$A\cos 4x+B\sin 4x$
	e^{5x}	Ae^{5x}
8	$(9x-2)e^{5x}$	$(Ax+B)e^{5x}$
9	x^2e^{5x}	$(Ax^2+Bx+C)e^{5x}$
10	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x+Be^{3x}\sin 4x$
11	$5x^2\sin 4x$	$(A_1x^2+B_1x+C_1)\cos 4x+(A_2x^2+B_2x+C_2)\sin 4x$
12	$xe^{3x}\cos 4x$	$(Ax+B)e^{3x}\cos 4x+(Cx+D)e^{3x}\sin 4x$

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:16 PM, 03 January 2022

Question # 2 of 10 (Start time: 07:17:25 PM, 03 January 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = x^2 e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	Ae^{5x}
○	$(Ax + B)e^{5x}$
○	$Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
●	$(Ax^2 + Bx + C)e^{5x}$

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Question # 3 of 10 (start time: 07:17:55 PM, 03 January 2022) Total Marks: 1

If a force acts upon a body, the acceleration is produced in the direction of the force whose magnitude is proportional to the magnitude of force, is known as the _____

Select the correct option

- (●) Newton's second law
- ( ) Newton's third law
- ( ) Newton's first law
- ( ) Hook's law

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 07:16 PM, 03 January 2022

Question # 4 of 10 (Start time: 07:20:13 PM, 03 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = (9x - 2)e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
○	Ae^{5x}
◉	$(Ax + B)e^{5x}$
○	$(Ax^2 + Bx + C)e^{5x}$

Click to Save Answer & Move to Next Question

For non - homogeneous differential equaitons, the form of the particular solution, $y_p = y_{p_1} + y_{p_2} + \ldots + y_{p_n}$, is a linear combination of all the linearly _____ functions generated by repeated differentiation of the input function $g(x)$.

Select the correct option

Reload Math Equations

- (●) independent
- ( ) dependent

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 07:16 PM, 03 January 2022

Question # 6 of 10 (Start time: 07:21:09 PM, 03 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) =$ constant (say 1), then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

◉	A
○	$Ax + B$
○	$Ax^2 + Bx + C$
○	$Ax^3 + Bx^2 + Cx + D$

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Question # 7 of 10 (Start time: 07:21:28 PM, 03 January 2022)

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

- (●) commute
- ( ) identical
- ( ) distributive
- ( ) inverse

Click

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BC190402733: SALMAN PERVAIZ

Time Left 58 sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:16 PM, 03 January 2022

Question # 8 of 10 (Start time: 07:21:39 PM, 03 January 2022)

Total Marks: 1

If

$$y = c_1 e^x + c_2 e^{4x}$$

is the complementary solution of

$$y'' - 5y' + 4y = 8e^x$$

, then the general form of its particular solution is

$$y_p$$

=_____.

Select the correct option

Reload Math Equations

- $(Ax + B)e^{4x}$
- Ae^x
- None of these
- Axe^x

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 07:23:44 PM, 03 January 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = xe^{3x} \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$(A_1 x^2 + B_1 x + C_1)\cos 4x + (A_2 x^2 + B_2 x + C_2)\sin 4x$
●	$(Ax + B)e^{3x}\cos 4x + (Cx + D)e^{3x}\sin 4x$
○	$Ae^{3x}\cos 4x + Be^{3x}\sin 4x$
○	$A\cos 4x + B\sin 4x$

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uestion # 1 of 10 (Start time: 06:46:28 PM, 03 January 2022)

In the method of variations of parameter, the derivative of the function, $u_2' =$ _____.

lect the correct option

○	$W \pm W_2$
○	$W.\ W_2$
○	$\frac{W}{W_2}$
◉	$\frac{W_2}{W}$

Click to Save An

The annihilator operator for the function, $g(x) = 5x$, is _____.

Select the correct option

Reload Math Equations

○	D
○	$D+5$
●	D^2
○	$D+1$

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 06:47:59 PM, 03 January 2022) **Total Marks:** 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = x^3 - x + 1$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$Ax + B$
○	A
○	$Ax^2 + Bx + C$
◉	$Ax^3 + Bx^2 + Cx + D$

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 06:46:55 PM, 03 January 2022)

Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____.

Select the correct option

◉	Displacement
○	Time
○	A spring constant
○	Mass

Click to Save Answer &

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 06:46 PM, 03 January 2022

Question # 5 of 10 (Start time: 06:48:36 PM, 03 January 2022)

Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \cos 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$(A + B)(\cos 4x + \sin 4x)$
○	Ae^{4x}
○	$Ax^2 + Bx + C$
◉	$A \cos 4x + B \sin 4x$

Click to Save Answer & Move to Next Question

MTH401 - Differential Equations (Quiz#2) Quiz Start Time:

Question # 6 of 10 (Start time: 06:48:53 PM, 03 January 2022)

Wronskian, $W(x, -3x) =$ _____.

Select the correct option

○	$-3x$
○	$3x$
○	1
◉	0

Click to Save Answer

Question # 10 of 10 (Start time: 06:52:21 PM, 03 January 2022)

What is annihilator operator of the function

$$g(x) = 8e^{3x}$$

?

Select the correct option

- (●) $(D-3)$
- ( ) $(D-3)^2$
- ( ) $(D+3)^2$
- ( ) $(D+3)$

Click to Save An

Question # 9 of 10 (Start time: 06:51:56 PM, 03 January 2022)

If

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option

○	Its annihilator operator is D^3
○	Its annihilator operator is D
◉	Its annihilator operator is D^2
○	Its annihilator operator is D+1

Question # 8 of 10 (Start time: 06:51:34 PM, 03 January 2022)

How are frequency and period related in simple harmonic motion?

Select the correct option

○	They are directly related
○	Their sum is constant
○	None of the above
◉	They are inversely related

Question # 1 of 10 (Start time: 10:07:13 AM, 04 January 2022)

The derivatives ${u_1}', {u_2}', \ldots, {u_n}'$ of the unknown functions $u_1, u_2, \ldots, u_n$ are find by using

Select the correct option

- ◉ ${u_n}' = \frac{W_k}{W}, \quad k = 1, 2, \ldots, n$
- ○ ${u_n}' = \frac{W}{W_k}, \quad k = 1, 2, \ldots, n$

Click to Save Ar

Question # 2 of 10 (start time: 10:08:00 AM, 04 January 2022) Total Marks: 1

The input function $g(x)$ can have one of the forms: A constant function k; A polynomial function; _____.

Select the correct option

Reload Math Equations

○	(III) Finite sums and products of these functions
○	(I) An exponential function $e^{\alpha x}$
○	(II) The trignometric functions $\sin \beta x$, $\cos \beta x$
◉	(IV) All (I), (II) and (III).

Click to Save Answer & Move to Next Question

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$(Ax^2 + Bx + C)e^{5x}$
○	Ae^{5x}
○	$(Ax + B)e^{5x}$
◉	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$

Click to Save Answer & Move to Next Question

The equation of free un-damped motion is _____.

Select the correct option

- $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$
- $\frac{d^2x}{dt^2} - \frac{k}{m}x = 0$

Click to Save Answ

Question # 5 of 10 (Start time: 10:08:49 AM, 04 January 2022) Total Marks: 1

For non - homogeneous differential equaitons, the form of the particular solution, $y_p = y_{p_1} + y_{p_2} + \ldots + y_{p_n}$, is a linear combination of all the linearly _____ functions generated by repeated differentiation of the input function $g(x)$.

Select the correct option

Reload Math Equations

- (●) independent
- ( ) dependent

Click to Save Answer & Move to Next Question

Vibration of an object on equilibrium point is called simple harmonic motion when the restoring force is proportional to _____.

Select the correct option

- A spring constant
- Mass
- Time
- Displacement

Click to Save Answer & Move to Next Qu

sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 10:07 AM, 04 January 2022

Question # 7 of 10 (start time: 10:09:16 AM, 04 January 2022)

Total Marks: 1

The periodic time is given by

Select the correct option

Reload Math Equations

◉	$\frac{2\pi}{\omega}$
○	$\frac{\pi}{\omega}$
○	$2\pi * \omega$
○	$\frac{\omega}{2\pi}$

Click to Save Answer & Move to Next Question

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

◉	$A\cos 4x + B\sin 4x$
○	$Ax^2 + Bx + C$
○	$(A + B)(\cos 4x + \sin 4x)$
○	Ae^{4x}

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 10:09:38 AM, 04 January 2022)

The differential operator D possesses the _____ property.

Select the correct option

- ○ inverse
- ○ nonlinearity
- ◉ linearity
- ○ closure

Click to Save Answer & Move to

MTH401 - Differential Equations (Quiz#2) Qu

Question # 10 of 10 (Start time: 10:10:08 AM, 04 January 2022)

In differential equations, the symbol D is known as _____ operator.

Select the correct option

- (●) differential
- ( ) relational
- ( ) integral
- ( ) arithmetic

Click to S

MTH401 - Differential Equations (Quiz#2) Quiz St

Question # 1 of 10 (Start time: 09:55:25 AM, 04 January 2022)

For a system in simple harmonic motion which of the following is the time required to complete a cycle of motion?

Select the correct option

- ○ Revolution
- ○ Frequency
- ○ Amplitude
- ◉ Period

Click to Save

Question # 2 of 10 (Start time: 09:55:59 AM, 04 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = x^3 - x + 1$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

◉	$Ax^3 + Bx^2 + Cx + D$
○	$Ax + B$
○	$Ax^2 + Bx + C$
○	A

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 09:56:22 AM, 04 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
○	$(Ax + B)e^{5x}$
○	$(Ax^2 + Bx + C)e^{5x}$
◉	Ae^{5x}

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 09:56:56 AM, 04 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) =$ constant (say 1), then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

◉	A
○	$Ax + B$
○	$Ax^3 + Bx^2 + Cx + D$
○	$Ax^2 + Bx + C$

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 09:56:38 AM, 04 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = x^2 e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
○	$(Ax + B)e^{5x}$
○	Ae^{5x}
●	$(Ax^2 + Bx + C)e^{5x}$

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 09:57:42 AM, 04 January 2022) Total Marks: 1

The annihilator operator for the function, $g(x) = 5x$, is _____.

Select the correct option

Reload Math Equations

○	D
○	$D+1$
○	$D+5$
◉	D^2

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 09:58:22 AM, 04 January 2022) **Total Marks:** 1

If we want to solve an initial value problem then we need to to apply the initial conditions to the _____ solution and not to the complementary function alone.

Select the correct option

- (●) general
- ( ) particular
- ( ) exact
- ( ) complementary

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 09:58:52 AM, 04 January 2022) Total Marks: 1

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$(Ax + B)e^{5x}$
○	$(Ax^2 + Bx + C)e^{5x}$
●	$Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
○	Ae^{5x}

Click to Save Answer & Move to Next Question

find:

e complementary function y_c, which is general solution of the associated omogeneous differential equation.
ny particular solution y_p of the non-homogeneous differential equation.

t the general solution of the non-homogeneous linear differential equation is given

General solution = Complementary function + Particular Integral

nding

Complementary function has been discussed in the previous lecture. In the next three ectures we will discuss methods for finding a particular integral for the non-homogeneous equation, namely

- The method of undetermined coefficients-*superposition approach*
- The method undetermined coefficients-*annihilator operator approach.*
- The method of variation of parameters.

The Method of Undetermined Coefficient

The method of undetermined coefficients developed here is limited to non-homogeneous linear differential equations

- That have constant coefficients, and
- Where the function $g(x)$ has a specific form

ned Coefficients-Superposition

ogeneous linear differential equation of order n is an equation of the

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = g(x)$$

fficients $a_0, a_1, \ldots, a_n$ can be functions of x. However, we will discuss
ns with constant coefficients.

to obtain the general solution of a non-homogeneous linear differential equation
must find:

The complementary function y_c, which is general solution of the associated homogeneous differential equation.

Any particular solution y_p of the non-homogeneous differential equation.

That the general solution of the non-homogeneous linear differential equation is given by

General solution = Complementary function + Particular Integral

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The form of $g(x)$

The input function $g(x)$ can have one of the following forms:

- A constant function k.
- A polynomial function
- An exponential function e^x
- The trigonometric functions $\sin(\beta x)$, $\cos(\beta x)$
- Finite sums and products of these functions.

Otherwise, we cannot apply the method of undetermined coefficients.

The method

Consist of performing the following steps.

Step 1 Determine the form of the input function $g(x)$.

Step 2 Assume the general form of y_p according to the form of $g(x)$

Step 3 Substitute in the given non-homogeneous differential equation.

Step 4 Simplify and equate coefficients of like terms from both sides.

Step 5 Solve the resulting equations to find the unknown coefficients.

Step 6 Substitute the calculated values of coefficients in assumed y_p

Restriction on g ?

The input function g is restricted to have one of the above stated forms because of th

reason:

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Trial particular solutions

Number	The input function $g(x)$	The assumed particular solution y_p
1	Any constant e.g. 1	A
2	$5x+7$	$Ax+B$
3	$3x^2-2$	Ax^2+Bx+c
4	x^3-x+1	Ax^3+Bx^2+Cx+D
5	$\sin 4x$	$A\cos 4x+B\sin 4x$
6	$\cos 4x$	$A\cos 4x+B\sin 4x$
7	e^{5x}	Ae^{5x}
8	$(9x-2)e^{5x}$	$(Ax+B)e^{5x}$
9	x^2e^{5x}	$(Ax^2+Bx+C)e^{5x}$
10	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x+Be^{3x}\sin 4x$
11	$5x^2\sin 4x$	$(A_1x^2+B_1x+C_1)\cos 4x+(A_2x^2+B_2x+C_2)\sin 4x$
12	$xe^{3x}\cos 4x$	$(Ax+B)e^{3x}\cos 4x+(Cx+D)e^{3x}\sin 4x$

If $g(x)$ equals a sum?

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ferential Operators

- In calculus, the differential coefficient d/dx is often denoted by the cap letter D. So that

$$\frac{dy}{dx} = Dy$$

The symbol D is known as differential operator.

- This operator transforms a differentiable function into another function, e.g.

$$D(e^{4x}) = 4e^{4x},\quad D(5x^3 - 6x^2) = 15x^2 - 12x,\quad D(\cos 2x) = -2\sin 2x$$

- The differential operator D possesses the property of linearity. This means tha f, g are two differentiable functions, then

$$D\{af(x) + bg(x)\} = aDf(x) + bDg(x)$$

Where a and b are constants. Because of this property, we say that D is a lin differential operator.

- Higher order derivatives can be expressed in terms of the operator D in a nat manner:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = D(Dy) = D^2y$$

Similarly $\frac{d^3y}{dx^3} = D^3y, \ldots, \frac{d^ny}{dx^n} = D^ny$

or one cycle.

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Period of Vibration

The simple harmonic motion of the suspended body is periodic and it repeats its position after a specific time period T. We know that the distance of the mass at any time t is given by

$$x = A\sin(\omega t + \phi)$$

$$A\sin\left[\omega\left(t + \frac{2\pi}{\omega}\right) + \phi\right]$$

$$= A\sin\left[(\omega t + \phi + 2\pi)\right]$$

$$= A\sin\left[(\omega t + \phi) + 2\pi\right]$$

Since

Therefore, the distances of the suspended body from the equilibrium position at the times t and $t + \frac{2\pi}{\omega}$ are same

ential Equations (MTH401)

erential Operators

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- Higher order derivatives can be expressed in terms of the operator D in a natural manner:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = D(Dy) = D^2y$$

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. So that $\frac{dy}{dx} = Dy$

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This operator transforms a differentiable function into another function, e.g.

$D(e^{4x}) = 4e^{4x}$, $D(5x^3 - 6x^2) = 15x^2$ [illegible] $D(\cos 2x) = -2\sin 2x$

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$D\{af(x) + bg(x)\} = aDf(x) + bDg(x)$

Where a and b are [illegible] because of this property, we say that D is a linear differential operator.

Higher order derivatives can be expressed in terms of the operator D in a natural manner: $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = D(Dy) = D^2y$

Trial particular solutions

umber	The input function $g(x)$	The assumed particular solution y_p
	Any constant e.g. 1	A
2	$5x+7$	$Ax+B$
3	$3x^2-2$	Ax^2+Bx+c
4		$Ax^3+Bx^$
5	$\sin 4x$	$A\cos 4x+B\sin 4x$
6		$A\cos 4x+B\sin 4x$
7	e^{5x}	Ae^{5x}
8	$(9x-2)e^{5x}$	$(Ax+B)e^{5x}$
9	x^2e^{5x}	$(Ax^2+Bx+C)e^{5x}$
10	$e^{3x}\sin 4x$	$Ae^{3x}\cos 4x+Be^{3x}$
11	$5x^2\sin 4x$	$(A_1x^2+B_1x+C$
12	$xe^{3x}\cos 4x$	$(Ax$

$\cos 4x+(A_2x^2+B_2x+C_2)\sin 4x$

$e^{3x}\cos 4x+(Cx+D)e^{3x}\sin 4x$

Trial particular solutions

Number	The input function $g(x)$	The assumed particular solution y_p
1	Any constant e.g. 1	A
2	$5x+7$	$Ax+B$
3	$3x^2-2$	Ax^2+Bx+c
4	x^3-x+1	Ax^3+Bx^2+Cx+D
5	$\sin 4x$	$A\cos 4x+B\sin 4x$
6	$\cos 4x$	$A\cos 4x+B\sin 4x$
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12	$xe^{3x}\cos 4x$	$(Ax+B)e^{3x}\cos 4x+(Cx+D)e^{3x}\sin 4x$

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ion

ionic motion of the suspended body is periodic and it repeats its position time period T. We know that the displacement of the mass at any time t is

$$x = A\sin(\omega t + \phi)$$

$$A\sin\left[\omega\left(t + \frac{2\pi}{\omega}\right) + \phi\right]$$

$$= A\sin[(\omega t + \phi + 2\pi)]$$

$$= A\sin[(\omega t + \phi)]$$

re, the distances of the suspended body from the equilibrium position at the and $t + \frac{2\pi}{\omega}$ are same

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uiz#2) Quiz

4:40 AM, 04 January 2022)

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____

D

$D + 5$

D^2

$D + 1$

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10:56 0.1 KB/s 48%

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

○	$(A_1 x^2 + B_1 x + C_1)\cos 4x + (A_2 x^2 + B_2 x + C_2)\sin 4x$
○	$(Ax + B)e^{3x}\cos 4x + (Cx + D)e^{3x}\sin 4x$
○	$Ae^{3x}\cos 4x + Be^{3x}\sin 4x$
○	$A\cos 4x + B\sin 4x$

Question # 6 of 10 (Start time: 11:23:25 AM, 04 January 2022) **Total Marks:** 1

The combination of the Newton's second law and the Hook's law could lead to a differential equation governing to the motion of a mass attached to spring i.e. _____ motion.

Select the correct option

- linear
- simple harmonic
- rotational
- translational

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$$y_c = c_1 \cos 2\sqrt{2}\,x + c_2 \sin 2\sqrt{2}\,x$$

Step 3 Since $D^2x = 0$, $(D+1)e^{-x} = 0$

Therefore the operators D^2 and $D+1$ annihilate the functions $5x$ and $12e^{-x}$. We apply $D^2(D+1)$ to the non-homogeneous differential equation

$$D^2(D+1)(D^2+8)y = 0.$$

This is a homogeneous differential equation of order 5.

Step 4 The auxiliary equation of this differential equation is

$$m^2(m+1)(m^2+8) = 0$$

$$\Rightarrow m = 0, 0, -1, \pm 2\sqrt{2}\,i$$

Therefore, the general solution of this equation must be

$$y = c_1 \cos 2\sqrt{2}x + c_2 \sin 2\sqrt{2}x + c_3 + c_4 x + c_5 e^{-x}$$

Step 5 Since the following terms are already present in y_c

$$c_1 \cos 2\sqrt{2}x + c_2 \sin 2\sqrt{2}x$$

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Lecture 22

Applications of Second Order Differential Equation

- A single differential equati… as mathematical mod… many different phenomena in science and …ngineering.
- Different forms of …e 2nd order linear differential equa… $a\frac{d^2y}{dx^2}+b\frac{dy}{dx}=f(x)$ appear in the analysis of problems in physics, chemistry and biology.
- In the present and next lecture we shall focus on one application; the motion of a mass attached to a spring.
- We shall see, what the individual terms $a\frac{d^2y}{dx^2}$, $b\frac{dy}{dx}$, cy and $f(x)$ means in the context of vibrational system.
- Except for the terminology and physical interpretation of the terms $a\frac{d^2y}{dx^2}$, $b\frac{dy}{dx}$, cy, $f(x)$ the mathematics of a series circuit is identical to that of a vibrating spring-mass system. Therefore we will discuss an *LRC* circuit in lecture.

Simple Harmonic Motion

When the Newton's 2nd law is combined with the Hook's Law, we can derive a differential equation governing the motion of a mass attached to spring–the simple harmonic motion.

Hook's Law

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mass attached to a spring.

- We shall see, what the individual terms $a\frac{d^2y}{dx^2}$, $b\frac{dy}{dx}$, cy and $f(x)$ means in the context of vibrational system.
- Except for the terminology and physical interpretation of the terms

$$a\frac{d^2y}{dx^2},\quad b\frac{dy}{dx},\quad cy,\quad f(x)$$

the mathematics of a series circuit is identical to that of a vibrating spring-mass system. Therefore we will discuss an *LRC* circuit in lecture.

22.1 Simple Harmonic Motion

When the Newton's 2nd law is combined with the Hook's Law, we can derive a differential equation governing the motion of a mass attached to spring–the simple harmonic motion.

22.1.1 Hook's Law

Suppose that

- A mass is attached to a flexible spring suspended from a rigid support, then
- The spring stretches by an amount 's'.
- The spring exerts a restoring F opposite to the direction of elongation or stretch.

The Hook's law states that the force F is proportional to the elongation s. i.e

$$F = ks$$

Where k is constant of proportionality, and is called spring constant.

Note That

- Different masses stretch a spring by different amount i.e s is different for different m.
- The spring is characterized by the spring constant k.
- For example if $W = 10$ lbs and $s = \frac{1}{2}ft$

  Then $F = ks$

  or $10 = \left(\frac{1}{2}\right)k$

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MTH401 - Differential Equations (Quiz#2) Quiz St

Question # 8 of 10 (Start time: 02:55:27 PM, 04 January 2022)

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

- ○ D+5
- ○ D+1
- ○ 5D
- ◉ D

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□ ○ ◁

3:36 PM

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BC190402733: SALMAN PERVAIZ

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 07:16

Question # 8 of 10 (Start time: 07:21:39 PM, 03 January 2022)

If

$$y = c_1 e^x + c_2 e^{4x}$$

is the complementary solution of

$$y'' - 5y' + 4y = 8e^x$$

, then the general form of its particular solution is

$$y_p$$

=_____.

Select the correct option

Relo

- (●) $(Ax + B)e^{4x}$
- ( ) Ae^x
- ( ) None of these
- ( ) Axe^x

Click to Save Answer & Move

5:53 PM
vulms.vu.edu.pk/Quiz/Q
45

BC190400820: NAGEEN AKHTAR

Time Left 82 sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 05:52 PM, 04 January 202

Question # 1 of 10 (Start time: 05:52:03 PM, 04 January 2022)

Total Marks:

The input function $g(x)$ can have one of the forms: A constant function k; A polynomial function; _____.

Select the correct option

Reload Math Equations

- (II) The trignometric functions $\sin \beta x$, $\cos \beta x$
- (III) Finite sums and products of these functions
- (IV) All (I), (II) and (III).
- (I) An exponential function $e^{\alpha x}$

Click to Save Answer & Move to Next Question

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MTH401 - Differential Equations (Quiz#2) Quiz Sta

Question # 2 of 10 (Start time: 05:52:14 PM, 04 January 2022)

The differential operator L is called an annihilator operator of the function f if: _____.

Select the correct option

- (ii) y = f(x) defines a sufficiently differentiable function.
- (iv) All (i), (ii) and (iii). ✓
- (iii) The function f is such that L(y)=0.
- (i) L is a linear differential operator with constant coefficients.

Click to Save

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BC190400820: NAGEEN AKHTAR

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 05:52

Question # 3 of 10 (Start time: 05:52:23 PM, 04 January 2022)

What is annihilator operator of the function

$$g(x) = x^4 + 5$$

?

Select the correct option

Rela

- ○ $(D+5)^4$
- ◉ D^5
- ○ (D+4)
- ○ (D-4)

Click to Save Answer & Move

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45

BC190400820: NAGEEN AKHTAR

Time Left 57 sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 05:52 PM, 04 January 2022

Question # 4 of 10 (Start time: 05:52:33 PM, 04 January 2022)

Total Marks: 1

Wronskian, $W(x, -3x) =$ _____.

Select the correct option

Reload Math Equations

- $-3x$
- 1
- $3x$
- 0

Click to Save Answer & Move to Next Question

5:55 PM
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45

Question # 5 of 10 (Start time: 05:53:24 PM, 04 January 2022)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x$
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____

Select the correct option

Reload M

- $(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$
- $(A_1x^2 + B_1x + C_1) \cos 4x + (A_2x^2 + B_2x + C_2) \sin 4x$
- $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
- $A \cos 4x + B \sin 4x$

Click to Save Answer & Move to N

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45

Question # 6 of 10 (Start time: 05:53:55 PM, 04 January 2022)

What is annihilator operator of the function

$$g(x) = 5$$

?

Select the correct option

- ○ D+1
- ○ D+5
- ○ 5D
- ◉ D

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45

BC190203902: IFRAHEEM HUSSAIN

MTH401 - Differential Equations (Quiz#2) Quiz Start

Question # 1 of 10 (Start time: 06:36:36 PM, 04 January 2022)

What is annihilator operator of the function

$$g(x) = 4\sin x$$

?

Select the correct option

◉	$(D^2 + 1)$
○	$(D^2 + 4)$
○	$(D^2 - 4)$
○	$(D^2 - 1)$

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BC190203902: IFRAHEEM HUSSAIN

MTH401 - Differential Equations (Quiz#2) Quiz Start Tir

Question # 2 of 10 (Start time: 06:37:22 PM, 04 January 2022)

The differential operator D possesses the _____ property.

Select the correct option

- nonlinearity
- closure
- linearity
- inverse

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BC190203902: IFRAHEEM HUSSAIN

Time Le

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 06:36 PM, 04 Jan

Question # 3 of 10 (Start time: 06:37:38 PM, 04 January 2022)

Tot

If a force acts upon a body, the acceleration is produced in the direction of the force whose magnitude is proportional to the magnitu force, is known as the _____.

Select the correct option

- ○ Hook's law
- ◉ Newton's second law
- ○ Newton's first law
- ○ Newton's third law

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□ ○ ◁

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BC190203902: IFRAHEEM HUSSAIN

MTH401 - Differential Equations (Quiz#2) Quiz

Question # 4 of 10 (Start time: 06:37:53 PM, 04 January 2022)

If

$$y = 2 + x$$

, then which of the following is true for it?

Select the correct option

- (●) Its annihilator operator is D^2
- ( ) Its annihilator operator is D^3
- ( ) Its annihilator operator is D
- ( ) Its annihilator operator is D+1

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BC190203902: IFRAHEEM HUSSAIN

Time Left 77 sec(s)

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 06:36 PM, 04 January 202

Question # 5 of 10 (Start time: 06:38:07 PM, 04 January 2022)

Total Marks

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$.
If the input function, $g(x) = e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Equations

- $(Ax^2 + Bx + C)e^{5x}$
- Ae^{5x}
- $(Ax + B)e^{5x}$
- $Ae^{5x}\cos 5x + Be^{5x}\sin 5x$

Click to Save Answer & Move to Next Question

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BC190203902: IFRAHEEM HUSSAIN

Time Le

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 06:36 PM, 04 Jan

Question # 6 of 10 (Start time: 06:38:31 PM, 04 January 2022)

To

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$. *If* the input function, $g(x) = x^2 e^{5x}$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Ed

- $Ae^{5x}\cos 5x + Be^{5x}\sin 5x$
- Ae^{5x}
- $(Ax + B)e^{5x}$
- $(Ax^2 + Bx + C)e^{5x}$

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BC190203902: IFRAHEEM HUSSAIN

MTH401 - Differential Equations (Quiz#2) Qu

Question # 7 of 10 (Start time: 06:39:18 PM, 04 January 2022)

The periodic time is given by

Select the correct option

◉	$\frac{2\pi}{\omega}$
○	$2\pi * \omega$
○	$\frac{\pi}{\omega}$
○	$\frac{\omega}{2\pi}$

Click to

6:42 PM

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BC190203902: IFRAHEEM HUSSAIN

MTH401 - Differential Equations (Quiz#2) Quiz Start

Question # 8 of 10 (Start time: 06:39:53 PM, 04 January 2022)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1$
If the input function, $g(x) = \cos 4x,$ then the assumed particular solution y_p could l

Select the correct option

○	Ae^{4x}
○	$Ax^2 + Bx + C$
●	$A\cos 4x + B\sin 4x$
○	$(A + B)(\cos 4x + \sin 4x)$

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BC190203902: IFRAHEEM HUSSAIN

Time Left

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 06:36 PM, 04 Janu

Question # 9 of 10 (Start time: 06:41:03 PM, 04 January 2022)

Tota

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$. *If* the input function, $g(x) = e^{3x} \sin 4x$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math Eq

- (●) $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
- ( ) $(Ax + B)e^{5x}$
- ( ) Ae^{5x}
- ( ) $(Ax^2 + Bx + C)e^{5x}$

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BC190203902: IFRAHEEM HUSSAIN

Time Left

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 06:36 PM, 04 Janu

Question # 10 of 10 (Start time: 06:42:06 PM, 04 January 2022)

Tota

The derivatives $u_1', u_2', \ldots, u_n'$ of the unknown functions $u_1, u_2, \ldots, u_n$ are find by using

Select the correct option

Reload Math Equ

- $u_n' = \frac{W}{W_k}, \quad k = 1, 2, \ldots, n$
- $u_n' = \frac{W_k}{W}, \quad k = 1, 2, \ldots, n$

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BC200200926: ARFAN SAJID

Time L

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 12:16 PM, 04 Ja

Question # 6 of 10 (Start time: 07:33:32 PM, 04 January 2022)

T

The velocity of a particle moving with simple harmonic motion is ---------- at the mean position.

Select the correct option

- Zero
- Maximum
- Minimum
- None of these

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□ ○ ◁

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45

BC200200926: ARFAN SAJID

MTH401 - Differential Equations (Quiz#2)

Question # 8 of 10 (Start time: 07:33:56 PM, 04 January 2022)

The hook's law states that the force F is proportional to the _____.

Select the correct option

- ○ None of these
- ○ Weight
- ◉ Elongation
- ○ Length

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BC190407427: WAQAS ALI

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 07:36 PM

Question # 1 of 10 (Start time: 07:36:31 PM, 04 January 2022)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g$
If the input function, $g(x) = 5x^2 \sin 4x$, then the assumed particular solution y_p could be _____

Select the correct option

Reloa

- $Ae^{3x} \cos 4x + Be^{3x} \sin 4x$
- $A \cos 4x + B \sin 4x$
- $(A_1 x^2 + B_1 x + C_1) \cos 4x + (A_2 x^2 + B_2 x + C_2) \sin 4x$
- $(Ax + B)e^{3x} \cos 4x + (Cx + D)e^{3x} \sin 4x$

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BC190407427: WAQAS ALI

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 07

Question # 2 of 10 (Start time: 07:36:43 PM, 04 January 2022)

The annihilator operator for the function, $g(x) = 2e^{-x}$, is _____.

Select the correct option

- ○ D
- ○ $D+2$
- ○ D^2
- ● $D+1$

Click to Save Answer & M

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Question # 3 of 10 (Start time: 07:36:57 PM, 04 January 2022)

The annihilator operator for the function, $g(x) = 3$, is _____.

Select the correct option

◉	D
○	$D+3$
○	$D+1$
○	$3D$

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BC190407427: WAQAS ALI

Time

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:36 PM, 04

Question # 4 of 10 (Start time: 07:37:07 PM, 04 January 2022)

For a second order linear ODE, if y_1 and y_2 are its solutions, and further $W(y_1, y_2) = -2$, $W_2 = \frac{e^x}{x}$, then $u'_2 =$ _____.

Select the correct option

Reload Mat

- $2x\,e^x$
- $\frac{e^x}{x}$
- $-\frac{e^x}{2x}$
- $\frac{e^{-x}}{2x}$

Click to Save Answer & Move to Next

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BC190407427: WAQAS ALI

MTH401 - Differential Equations (Quiz#2)

Question # 5 of 10 (Start time: 07:37:24 PM, 04 January 2022)

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

- ○ inverse
- ○ identical
- ◉ commute
- ○ distributive

Cli

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BC190409260: ABDUL QAYYUM

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:56 PM,

Question # 1 of 10 (Start time: 07:56:51 PM, 04 January 2022)

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g($
If the input function, $g(x) = 5x + 7$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload

- $Ax^2 + Bx + C$
- $Ax + B$
- $Ax^3 + Bx^2 + Cx + D$
- A

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BC190409260: ABDUL QAYYUM

MTH401 - Differential Equations (Quiz#2) Quiz Start Time: 07

Question # 2 of 10 (Start time: 07:57:10 PM, 04 January 2022)

The derivatives $u_1', u_2', \ldots, u_n'$ of the unknown functions $u_1, u_2, \ldots, u_n$ are find by using

Select the correct option

- (●) $u_n' = \frac{W_k}{W}, \quad k = 1, 2, \ldots, n$
- ( ) $u_n' = \frac{W}{W_k}, \quad k = 1, 2, \ldots, n$

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BC190409260: ABDUL QAYYUM

MTH401 - Differential Equations (Quiz#2)

Question # 3 of 10 (Start time: 07:57:24 PM, 04 January 2022)

The gravitational force exerted by the earth on a body of mass m is called ---------- of the bo

Select the correct option

- (●) weight
- ( ) Force

7:59 PM
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MTH401 - Differential Equations (Quiz#2)

Question # 4 of 10 (Start time: 07:57:34 PM, 04 January 2022)

How are frequency and period related in simple harmonic motion?

Select the correct option

- ○ None of the above
- ○ They are directly related
- ○ Their sum is constant
- ◉ They are inversely related

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BC190409260: ABDUL QAYYUM

MTH401 - Differential Equations (Quiz#2)

Question # 5 of 10 (Start time: 07:57:46 PM, 04 January 2022)

What is annihilator operator of the function

$$g(x) = x^4 + 5$$

?

Select the correct option

◉	D^5
○	$(D+5)^4$
○	(D+4)
○	(D-4)

7:59 PM
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BC190409260: ABDUL QAYYUM

MTH401 - Differential Equations (Quiz#2)

Question # 6 of 10 (Start time: 07:58:01 PM, 04 January 2022)

The differential operator L is called an annihilator operator of the function f if: _____.

Select the correct option

- (iii) The function f is such that L(y)=0.
- (i) L is a linear differential operator with constant coefficients.
- (iv) All (i), (ii) and (iii). ●
- (ii) y = f(x) defines a sufficiently differentiable function.

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BC190409260: ABDUL QAYYUM

Time L

MTH401 - Differential Equations (Quiz#2)

Quiz Start Time: 07:56 PM, 04 Ja

Question # 7 of 10 (Start time: 07:58:13 PM, 04 January 2022)

To

Consider the non - homogeneous linear differential equation, $a_n y^{(n)} + a_{n-1} y^{(n-1)} + \ldots + a_1 y' + a_0 y = g(x)$. If the input function, $g(x) = 3x^2 - 2$, then the assumed particular solution y_p could be _____.

Select the correct option

Reload Math E

- (●) $Ax^2 + Bx + C$
- ( ) A
- ( ) $Ax^3 + Bx^2 + Cx + D$
- ( ) $Ax + B$

Click to Save Answer & Move to Next Q

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BC190409260: ABDUL QAYYUM

MTH401 - Differential Equations (Quiz#2) Quiz Star

Question # 8 of 10 (Start time: 07:58:30 PM, 04 January 2022)

If

$$y = c_1 e^{(-2+\sqrt{6})x} + c_2 e^{(-2-\sqrt{6})x}$$

is the complementary solution of

$$y'' + 4y' - 2y = 2x^2 - 3x + 6$$

,then the general form of its particular solution is

$$y_p$$

=_____.

Select the correct option

- $Ax^3 + Bx^2 + Cx + D$
- $Ax + B$
- $(Ax^2 + Bx)2x^2$
- $Ax^2 + Bx + C$

Click to Save A

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BC190409260: ABDUL QAYYUM

MTH401 - Differential Equations (Quiz#2) Qui

Question # 9 of 10 (Start time: 07:58:48 PM, 04 January 2022)

Wronskian of the functions, $y_1(x) = 1,\ y_2(x) = \sin x,\ y_3(x) = \cos x$, is _____.

Select the correct option

◉	1
○	2
○	-1
○	0

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MTH401 - Differential Equations (Quiz#2)

Question # 10 of 10 (Start time: 07:58:58 PM, 04 January 2022)

The polynomial linear differential operators with constant coefficients _____.

Select the correct option

- ○ distributive
- ○ inverse
- ◉ commute
- ○ identical